Reg. No : 2 1 1 6 2 3 1 0 0 1 2 0 3

Question Paper Code :151101

B.E./B.Tech DEGREE EXAMINATION NOV/DEC - 2025

Fifth Semester

Information Technology

AI23231 PRINCIPLES OF ARTIFICIAL INTELLIGENCE

[Common to: AIML, AIDS and CSD]

(Regulations 2023)

Time: Three Hours Maximum : 100 Marks

Answer ALL Questions

PART A (10x2=20 Marks)

1	Define Artificial Intelligence. List any two key problems in AI research.	[CO1]	[K1]
2	Classify the different types of intelligent agents based on their learning ability.	[CO1]	[K2]
3	List any two differences between Breadth-First Search and Depth-First Search.	[CO2]	[K1]
4	What is a heuristic function in A* search, and why is it used?	[CO2]	[K2]
5	Define constraint satisfaction problem. Give one real-life example.	[CO3]	[K1]
6	What is the purpose of alpha-beta pruning in game trees?	[CO3]	[K1
7	What does Bayesian theorem state? Mention one AI application that uses it.	[CO4]	[K1]
8	Differentiate between procedural and declarative knowledge.	[CO4]	[K2]
9	List any two main categories of Machine Learning.	[CO5]	[K1]
10	Define classification and clustering with one key difference.	[CO5]	[K2]

PART B (5x13= 65 Marks)

11.	a.	Illustrate the structure of an intelligent agent with a neat labelled diagram. Discuss the roles of sensors, effectors, and performance measures. (Include Agent Architecture Diagram).	[CO1]	[K3]
		[OR]		
	b.	Explain the Tic-Tac-Toe problem as a state-space search. Draw and label the search tree up to level three showing possible moves of both players.	[CO1]	[K3]

12.	a.	Elaborate the working of Breadth-First Search and Depth-First Search algorithms. Compare their space and time complexities with a diagrammatic representation.	[CO2]	[K2]
		[OR]		
	b.	Explain the concept of Heuristic Search using A and AO* algorithms. Draw the corresponding search graph for any sample problem.	[CO2]	[K2]
13.	a.	Describe how Constraint Satisfaction Problems (CSP) are solved using local search methods. Illustrate the working with an example of a 4-Queens problem. (Include CSP Search Diagram).	[CO3]	[K2]
		[OR]		
	b.	Draw and explain the Minimax Search Tree and demonstrate how alpha-beta pruning reduces the number of nodes evaluated.	[CO3]	[K2]
14.	a.	Explain Bayesian Networks and their role in probabilistic reasoning. Construct and label a simple Bayesian Network for a disease diagnosis problem.	[CO4]	[K2]
		[OR]		
	b.	Compare Forward and Backward Reasoning approaches in AI. Draw the corresponding inference diagrams for each method.	[CO4]	[K2]
15.	a.	Discuss various types of Machine Learning Paradigms (Supervised, Unsupervised, and Reinforcement). Illustrate with an example diagram showing the learning workflow.	[CO5]	[K2]
		[OR]		
	b.	Describe the architecture of a simple Neural Network. Explain how it performs classification tasks.	[CO5]	[K2]

PART C (1x15=15 Marks)

16.	a.	Design an AI-based robotic assistant system that can learn from user commands and adapt its environment dynamically. Explain the architecture, agent design, and learning approach used.		[K5]
		[OR]		
	b.	Propose a Machine Learning framework for detecting fraudulent financial transactions using classification and probabilistic reasoning. Explain the data flow and learning components.		[K5]

Reg. No : 2 1 1 6 2 4 1 5 0 1 0 8 0

Question Paper Code :031181

B.E./B.Tech DEGREE EXAMINATION APRIL/MAY - 2025

Second Semester

Artificial Intelligence and Data Science

AI23231 PRINCIPLES OF ARTIFICIAL INTELLIGENCE

(Common to: Artificial Intelligence and Machine Learning)

(Regulations 2023)

Time : Three Hours | Maximum : 100 Marks

Answer ALL Questions

PART A (10x2=20 Marks)

1	Define intelligent agent with an example.	[CO1]	[A1]
2	List any two characteristics of problem-solving agents.	[CO1]	[A1]
3	Differentiate between breadth-first search and depth-first search.	[CO2]	[A2]
4	What is the heuristic function in A* search?	[CO2]	[A2]
5	Define constraint satisfaction problem.	[CO3]	[A1]
6	Write the purpose of alpha-beta pruning in adversarial search.	[CO3]	[A1]
7	State the difference between procedural and declarative knowledge.	[CO4]	[A2]
8	What is logic programming in AI?	[CO4]	[A1]
9	Write a short note on Bayesian networks.	[CO5]	[A1]
10	Define certainty factors in expert systems.	[CO5]	[A1]

PART B (5x13= 65 Marks)

11	a.	Explain the structure of agents and differentiate between goal-based and utility-based agents.	[CO1]	[B1]
		OR		
	b.	Describe the components and characteristics of a problem in state space search.	[CO1]	[B1]

12 a.	Apply the A* search algorithm to solve a sample path-finding problem with given heuristics.	[CO2]	[B2]
	OR		
b.	Apply simulated annealing to solve a local optimization problem and explain its advantages.	[CO2]	[B2]
13 a.	Analyze how the min-max algorithm works in game theory with an example game tree.	[CO3]	[B2]
	OR		
b.	Analyze the process of alpha-beta pruning and demonstrate how it improves efficiency.	[CO3]	[B2]
14 a.	Evaluate the significance of rule-based knowledge representation in expert systems.	[CO4]	[B1]
	OR		
b.	Evaluate the forward and backward reasoning techniques in AI with an illustrative example.	[CO4]	[B1]
15 a	Apply Bayesian networks and fuzzy logic in reasoning under uncertainty with a sample case.	[CO5]	[B1]
	OR		
15 b	Demonstrate the working of Dempster-Shafer theory and explain its use in expert systems.	[CO5]	[B1]

PART C (1x15=15 Marks)

16 a.	Create an intelligent agent framework for a real-time environment. (e.g., smart traffic control).	[C1]	
	OR		
b.	Design an AI-based decision-making system using fuzzy logic and Bayesian reasoning.	[C1]	

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Question Paper Code :201151

B.E./B.Tech DEGREE EXAMINATION APRIL/MAY- 2024

Second Semester

Artificial Intelligence and Data Science

AI23231 PRINCIPLES OF ARTIFICIAL INTELLIGENCE

(Common to AIML)

(Regulations 2023)

Time : Three Hours | Maximum : 100 Marks

Answer ALL Questions

PART A (10x2=20 Marks)

1.	Analyze the measure of the performance of an agent.	[CO1]	[B2]
2.	What are utility based agents?	[CO1]	[A1]
3.	What is the power of heuristic search?	[CO2]	[A1]
4.	Distinguish between Simple hill climbing and Simulated annealing.	[CO2]	[B2]
5.	What do you mean by Constraint Satisfaction?	[CO3]	[A1]
6.	Define pruning.	[CO3]	[A1]
7.	What is rule based knowledge representation?	[CO4]	[A1]
8.	Define Declarative and procedural knowledge.	[CO4]	[A1]
9.	Define Dempster-Shafter Theory.	[CO5]	[A1]
10.	What is a Bayesian network?	[CO5]	[A1]

PART B (5x13= 65 Marks)

11.	a.	Discuss in detail about the structure of agents in Artificial Intelligence.	[CO1]	[B1]

[OR]

	b.	How to define a problem as state space search? Discuss it with the help of an example.	[CO1]	[A1]

12. a. Explain AO* algorithm. Give one example where AO* is suitable to apply. [CO2]

[OR]

b. Discuss the following search Technique with the help of an example. Also discuss the benefits and shortcoming of each [CO2]

i) Breadth First Search.

ii) Depth First Search.

13. a. Develop a Game tree with the steps involved for the depth 3 and branching factor 3 using Alpha-Beta Pruning algorithm. [CO3]

[OR]

b. Explain the min max search procedure with example. [CO3]

14. a. Discuss about the AI for knowledge representation with suitable example. [CO4]

[OR]

b. Elaborate the concept of the following, [CO4]

(i) Forward Reasoning.

(ii) Backward Reasoning.

15. a. Discuss briefly about Bayesian Networks with example. [CO5]

[OR]

b. Illustrate the Markov decision process with neat diagram. [CO5]

PART C (1x15=15 Marks)

16. a. Evaluate Greedy Best First Search with an example. [C1]

[OR]

b. Illustrate detailed note Rule based system architecture for expert system. [C2]